AQUAFAB INDUSTRIES

PIV Flow Measurement and Chamber Velocity Calibration

Setting up the camera, measuring a flow field, and calibrating pump output to chamber velocity.

For Aquafab Workbench 0.1.0

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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1 PIV flow measurement and chamber velocity calibration

Particle image velocimetry (PIV) measures how the water itself is moving, by following seeded tracer particles from one video frame to the next. Aquafab Workbench uses it two ways:

- **Flow Field (PIV)**, under Experiments: measure the velocity field in a clip or live camera view. To check the working section is uniform, find dead zones, or document a flow. Needs a license.
- **Chamber Velocity (PIV)**, under Calibrations: drive the pumps through a sequence and measure the chamber velocity at each setting, so the machine's flow control reads in cm/s instead of pump percent. Does not need a license.

> Screenshots use a synthetic seeded-flow video made for this guide. The velocities in them are not measurements.

1.1 Part 1: The camera and the water

Good PIV starts at the camera, not in the settings.

1. **Seed the water** with neutrally buoyant tracer particles, densely enough that every 32 × 32-pixel window holds several.
2. **Light it evenly**, ideally a thin sheet across the flow. Avoid glare and reflections from walls; they correlate as particles that do not move.
3. **Fix the camera** square to the working section, so the scale is the same across the frame. A USB camera on a rigid mount is enough.
4. **Choose the frame rate** so particles move a few pixels between frames, no more than about a quarter of the window size. Faster flow needs a higher frame rate.
5. **Put a known distance in view**, in the plane of the flow: a ruler, or the chamber's own width.

1.1.1 Connecting the camera

The controls over the picture are the same on every video page:

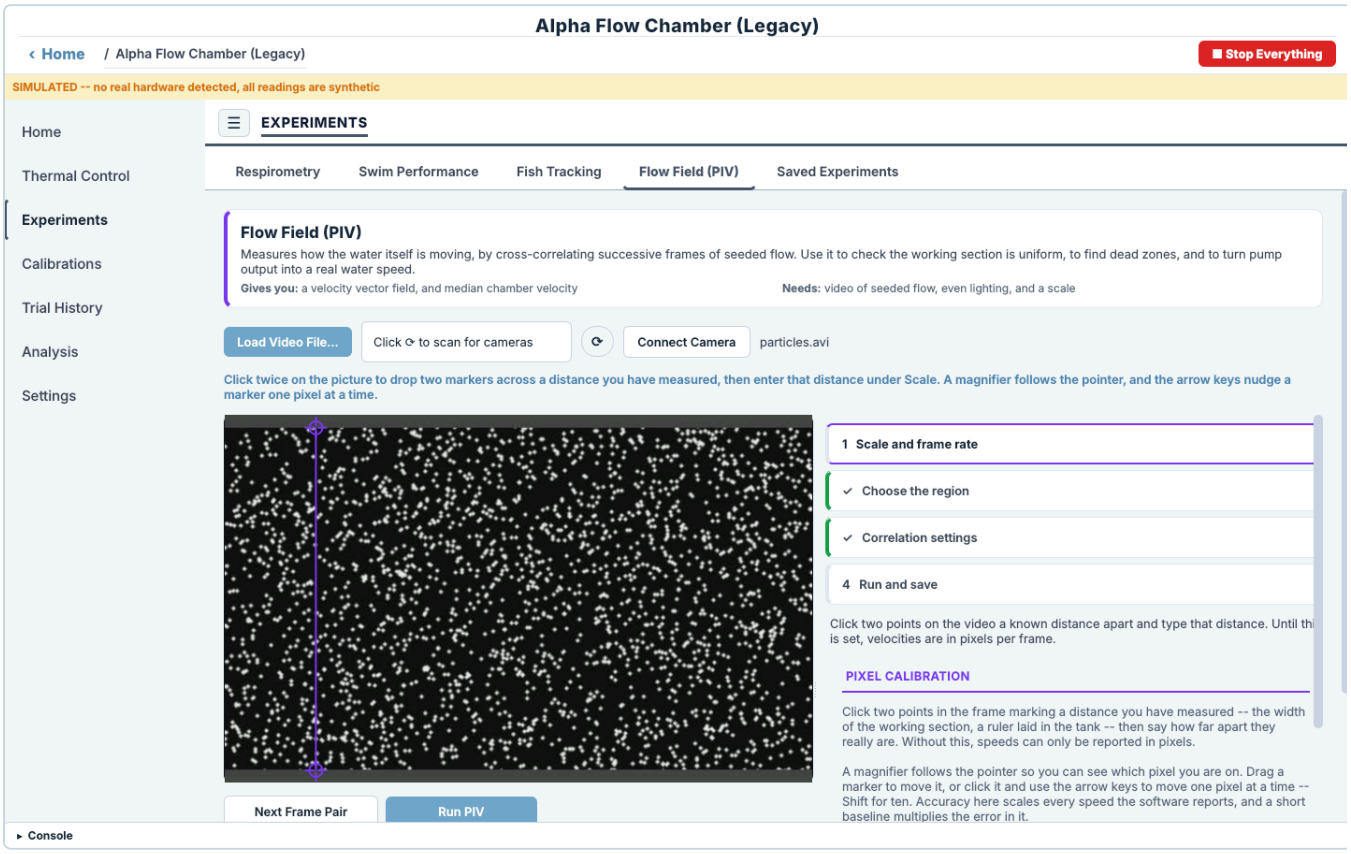
Control	What it does
Load Video File…	Opens a clip (`.avi`, `.mp4`, `.mov`, `.mkv`, `.m4v`).
Camera list, ⟳	The cameras this computer sees; ⟳ rescans after plugging one in.
Connect Camera	Starts the live picture.
Pause / **Disconnect**	Freezes the picture, or releases the camera for another page.

Only one page can use a camera at a time.

1.2 Part 2: Measuring a flow field

Experiments → Flow Field (PIV) (or **Video Analysis** on the Launcher, for footage from anywhere). The stages are in a strip on the right.

1.2.1 1. Scale and frame rate

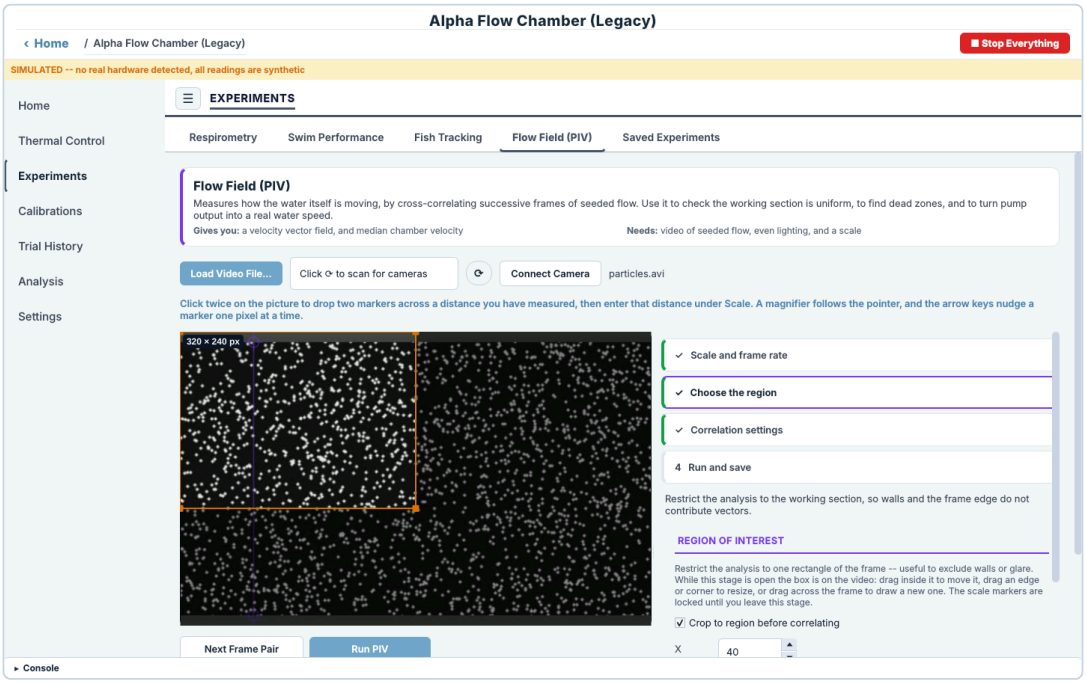


Placing the scale markers.

1. Click once at one end of the known distance, and again at the other. A magnifier shows the exact pixel; drag a marker, or select it and use the arrow keys (Shift for ten pixels) to fine-tune.
2. Type **Distance between markers** (cm) and press **Apply scale from markers**.
3. Leave **Frame rate** at 0 unless the camera reports it wrongly; every velocity scales with it.

Use the longest distance you can. Accuracy here scales every velocity.

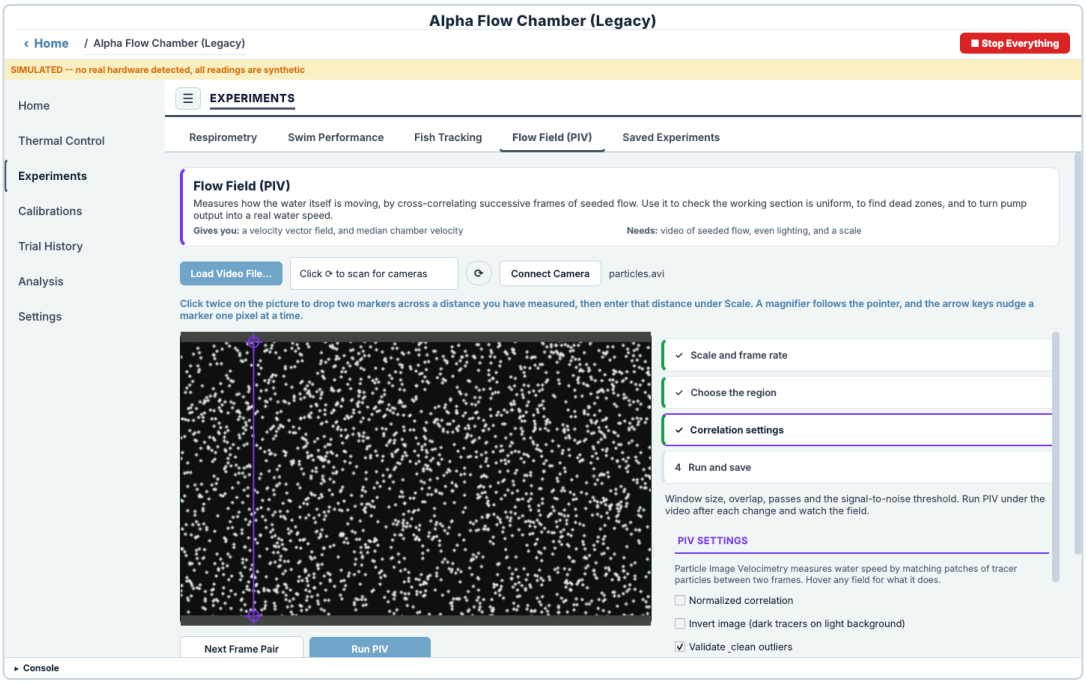
1.2.2 2. Choose the region



The region box.

Drag a box over the working section, away from walls and glare (drag inside to move, edges to resize), and tick **Crop to region before correlating**.

1.2.3 3. Correlation settings

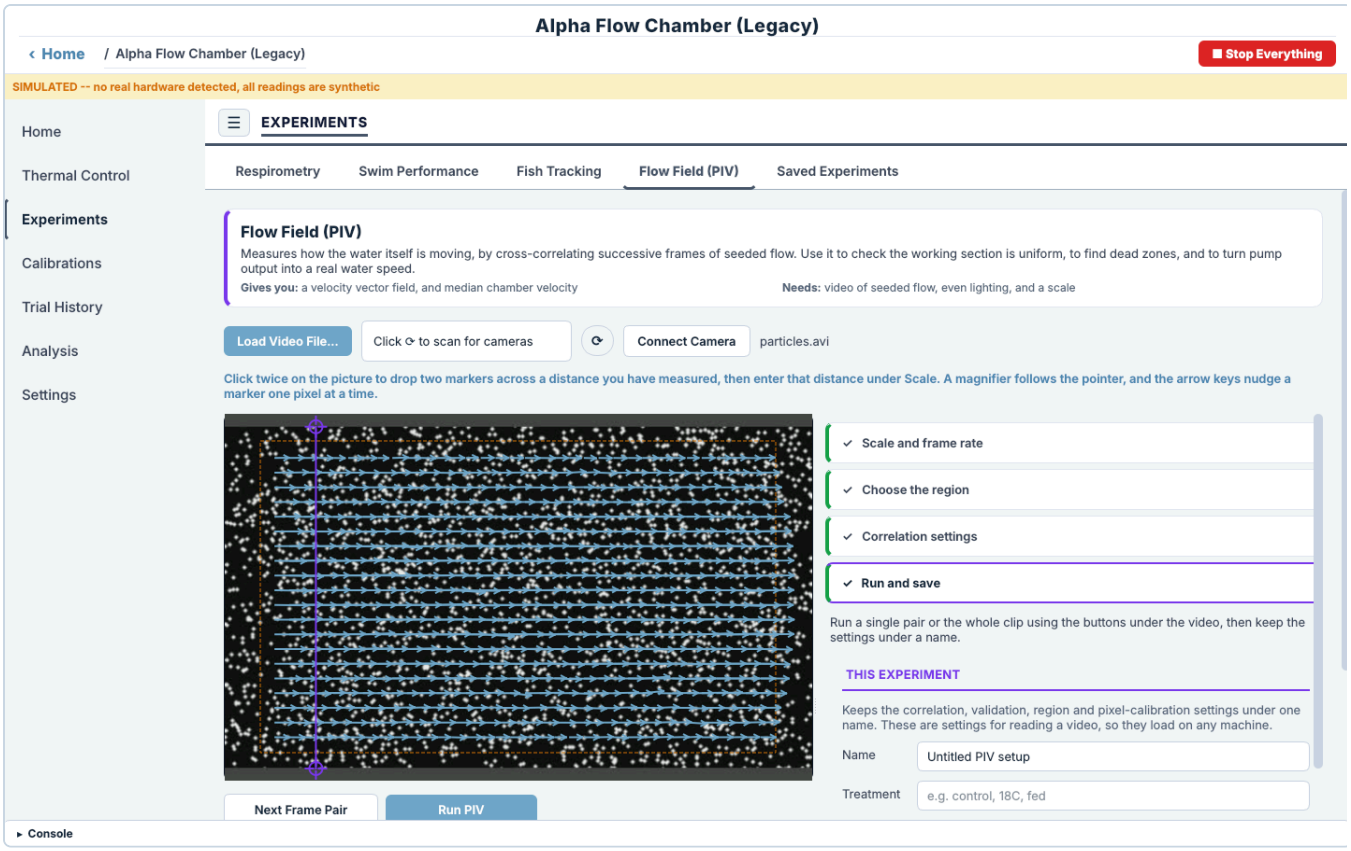


PIV settings.

Start with the defaults: **Window size** 32 px, **Overlap** 16 px, **Passes** 1, *circular* correlation, *gaussian* subpixel, **Validate & clean outliers** on. If many vectors are flagged: try a larger window, 2–3 passes, **Normalized correlation** for uneven lighting, or **Invert image** for dark particles on a light background. Hover any field for what it does.

1.2.4 4. Run and save

Button	What it does
Run PIV	Analyses the current frame pair and draws the vectors. Quick, for trying settings.
Next Frame Pair	Steps forward one frame.
Analyze Whole Video	Analyses the clip pair by pair, charts the median u and v over time, and saves the run to Trial History.
Export Vector Field…	Saves the vectors as CSV.



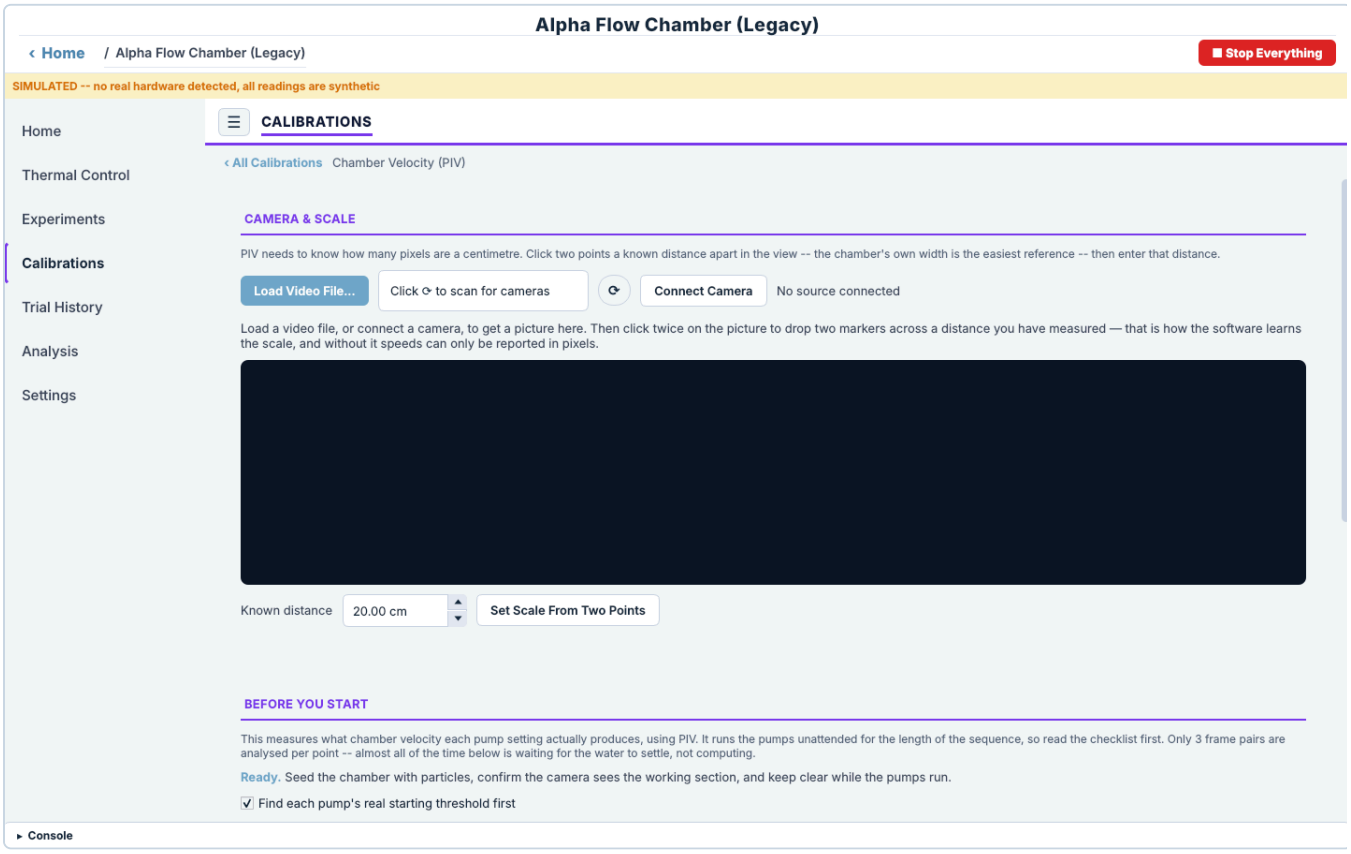
A vector field and the median velocity over the clip.

The **Result** card gives the median flow in px/frame and, with a scale, in cm/s, and how many vectors were flagged and cleaned. *u* is along the frame's horizontal axis, *v* along its vertical.

1.3 Part 3: Calibrating chamber velocity

This teaches the machine what chamber velocity each pump setting produces. Afterwards, the Home page's **Chamber flow** slider reads in cm/s, a water speed in a trial can be set in cm/s, and a multi-pump machine coordinates its pumps to follow the slider smoothly.

> It runs the pumps unattended through a sequence. **No animal in the chamber**, and keep clear while it runs.



The Chamber Velocity calibration.

1. Fill the machine, seed the water, and set up the camera as in Part 1.
2. Check **Settings → This machine's limits → Working section cross-section** is entered (it is used to cross-check against the flow meter).
3. **Calibrations → Chamber Velocity (PIV) → Open.**
4. **Camera & scale**: **Connect Camera**; click two points a known distance apart; type **Known distance**; press **Set Scale From Two Points**.
5. **Before you start** must say **Ready**. If it says **Not ready**, it lists exactly what is missing.
6. Leave **Find each pump's real starting threshold first** ticked. It finds where each pump actually starts moving water, rather than trusting the figure in the profile.
7. Note the plan (for example *"18 measurements, about 19 minutes"*) and press **Start Calibration**. Each point settles (*"Settling — 40s"*), then is measured (*"Measuring with PIV..."*). **Abort & Stop Pumps** stops at once and saves nothing.

8. When it reads *Calibration complete*, review **Measured points**: **Pump settings**, **Velocity**, **Implied flow** and the flow **Meter**. A velocity of zero with a pump running needs looking into.
9. Press **Save to Profile**.

Why PIV rather than the flow meter: the meter sees only one leg of the loop, not the water the animal is in, and where two pumps feed the chamber they interact, so one running harder can even slow the chamber. PIV measures the chamber directly, over a grid of pump combinations.

Run the calibration again after any change to pumps, plumbing or the working section.